Problem 7.7: A fluid is pumped from a pressurized tank with a pump that is placed 4.5 m higher elevation from the fluid surface of the pressurized tank to an open tank 3 m below the fluid surface of the pressurized tank. If the pump adds 1.5 kW to the system determine the discharge and the pressure at point A. If the vaporization pressure of the fluid is 3 N/cm^2 (absolute pressure), show if the fluid evaporate or not at this point.

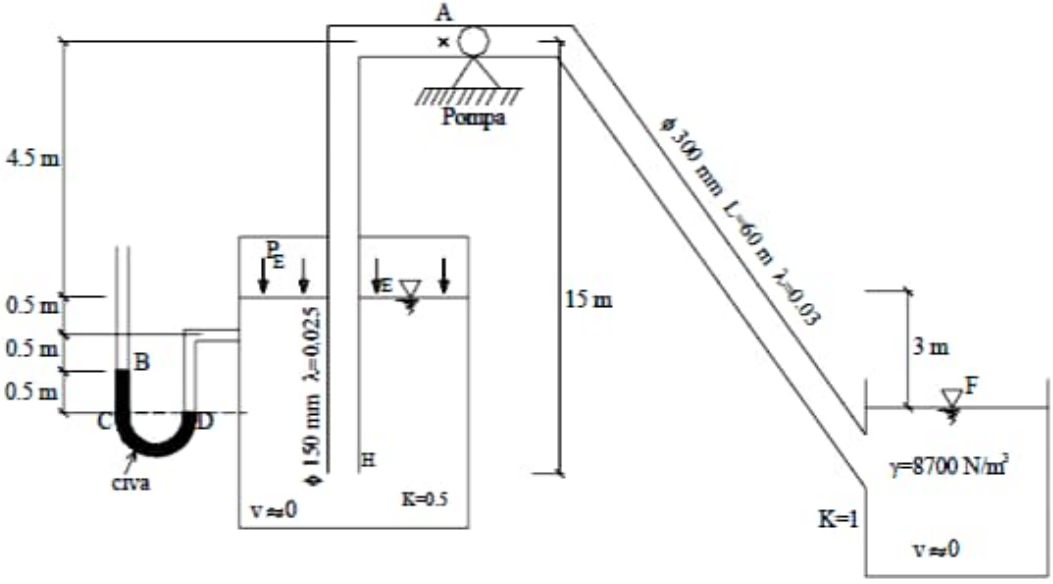


Figure 7.7

Solution:

First of all find the pressure p_E by using manometer.

$p_C = p_D$

$0.5 \times 136000 - 1.5 \times 8700 + p_E$

$P_E = 54950\ N/m^2$

The energy equation between E-F

$$z_E + \frac{V_E^2}{2g} + \frac{P_E}{\gamma} + H_m = z_F + \frac{V_F^2}{2g} + \frac{P_F}{\gamma} + \sum \Delta H$$

$$\sum \Delta H = \Delta h_g + \Delta h_{L_{HA}} + \Delta h_{L_{AE}} + \Delta h_C$$

$$\sum \Delta H = 0.5 \times \frac{V_{AH}^2}{2g} + 0.025 \times \frac{15}{0.15} \times \frac{V_{AH}^2}{2g} + 0.03 \times \frac{60}{0.30} \times \frac{V_{AF}^2}{2g} + \frac{V_{AF}^2}{2g}$$

Continuity equation

$Q = V_{AH} \times A_{AH} = V_{AF} \times A_{AF}$

$$V_{AH} = \frac{A_{AG}}{A_{AH}} V_{AG}$$

$V_{AH}=4\ V_{AF}$

$$\sum \Delta H = 0.5 \times \frac{(4V_{AF})^2}{2g} + 0.025 \times \frac{15}{0.15} \times \frac{(4V_{AF})^2}{2g} + 0.03 \times \frac{60}{0.30} \times \frac{V_{AF}^2}{2g} + \frac{V_{AF}^2}{2g}$$

$$\sum \Delta H = 2.80 V_{AF}^2$$

$$h = \frac{P_E}{\gamma} = \frac{P_G}{\gamma} = \frac{54950}{8700} = 6.32\,m$$

$3+6.32+H_m=2.80\ V_{AF}^2$

The pump power

$$N = \frac{\gamma\ Q\ H_m}{1}$$

$$15 = \frac{8700 \times 10^{-3} \times Q \times H_m}{1}$$

$$H_m = \frac{0.172}{Q}$$

If this is substituted into the energy equation

$$9.32 + \frac{0.172}{Q} = 2.80 \times \left(\frac{Q}{\frac{\pi \times 0.3^2}{4}} \right)^2$$

$560.39\,Q^3 - 9.32\,Q - 0.172 = 0$

By trial and error the discharge is obtained as $Q=0.137\ m^3/s$.

Let's write the energy equation between E-A to find the pressure at A.

$$z_E + \frac{V_E^2}{2g} + \frac{P_E}{\gamma} = z_A + \frac{V_A^2}{2g} + \frac{P_A}{\gamma} + 0.5 \times \frac{V_{AH}^2}{2g} + 0.025 \times \frac{15}{0.15} \times \frac{V_{AH}^2}{2g}$$

$$V_{AH} = \frac{0.137}{\frac{\pi \times 0.15^2}{4}} = 7.75\ m/s$$

$$6{,}32 = 4{,}5 + \frac{P_A}{\gamma} + 3 * \frac{V_{AH}^2}{2g} \quad ; \qquad \frac{P_A}{\gamma} = -7.36m$$

$$V_{AH} = \frac{0.137}{\frac{\pi \times 0.15^2}{4}} = 7.75\,\text{m/s}$$

$$6{,}32 = 4{,}5 + \frac{P_A}{\gamma} + 3 * \frac{V_{AH}^2}{2g} \quad ; \qquad \frac{P_A}{\gamma} = -7.36m$$

p_A=-7.36×8700=-64032 N/m^2 (Relative pressure)

$(p_A)_{ABSOLUTE}=p_{ATM}+(p_A)_{REL}$=100000 (N/m^2)+(-64032)=35968 N/m^2

$(p_A)_{ABSOLUTE}$=3.6 N/cm^2 >3 N/cm^2 fluid doesn't evaporate at A.

Problem 7.8: The discharge from the reservoir A to reservoir C is 400 lt/s in Figure 7.8. The pressure at the outlet of the turbine DE is −5 mwc. Find the power which could be obtained from the turbine and draw the energy grade line of the system.

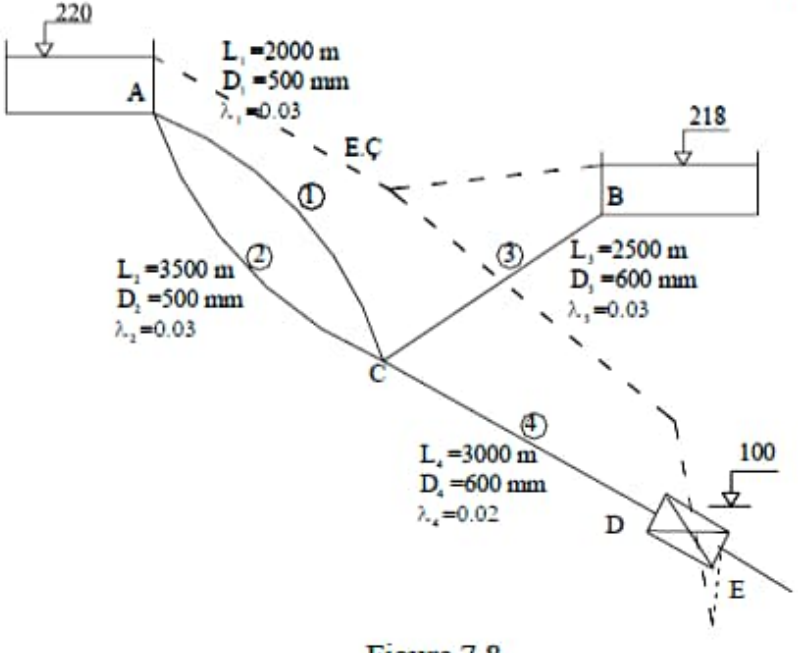


Figure 7.8

Solution:

For the **parallel pipes** (1) and (2)

Continuity equation $Q_1+Q_2=0.4\ m^3/s$

From the friction loss equality $\Delta h_{L1}=\Delta h_{L2}$

$$f_1\frac{L_1}{D_1}\frac{V_1^2}{2g}=f_2\frac{L_2}{D_2}\frac{V_2^2}{2g}$$

$$L_1\left(\frac{Q_1^2\times 16}{\pi\times D_1^4}\right)=L_2\left(\frac{Q_2^2\times 16}{\pi\times D_2^4}\right)$$

$$L_1Q_1^2=L_2Q_2^2$$

$$Q_1=\sqrt{\frac{3500}{2000}}Q_2=1.3Q_2$$

$$Q_1 = \sqrt{\frac{3500}{2000}} Q_2 = 1.3 Q_2$$

$$Q_1 + Q_2 = 1.3Q_2 + Q_2 = 0.4$$

Q_2=0.174 m^3/s V_2=0.887 m/s

Q_1=0.266 m^3/s V_1=1.15 m/s

$$\Delta h_{L1} = 0.03 \times \frac{2000}{0.5} \times \frac{1.15^2}{19.67} = 8.11\,\text{m}$$

The piezometer height at C

$$\left(\frac{P}{\gamma} + z\right)_C = 220 - 8.11 = 211.89\,\text{m}$$

Δh_{L3}=218-211.89=6.11 m

$$6.11 = 0.03 \frac{2500}{0.6} \frac{V_3^2}{19.62}$$ V_3=0.979 Q_3=0.280 m^3/s

The discharge of pipe 4is obtained from continuity equation

Q_4=0.4+0.28=0.68 m^3/s

V_4=2.4 m/s

$$\Delta h_{L4}=0.02\times\frac{3000}{0.6}\times\frac{2.4^2}{19.62}=29.36\,\text{m}$$

The piezometer height at D

$$\left(\frac{P}{\gamma}+z\right)_0=211.89-29.36=182.53\ \text{m}\qquad \text{ve}\ \left(\frac{P}{\gamma}+z\right)_0=182.53-100=82.53\ \text{m}$$

the manometric height is obtained as H_m=82.53+5=87.53 m

$$N = \frac{\gamma(kN/m^3) \times Q(m^3/s) \times H_m(m)}{\eta} (kW)$$

$$N = \frac{(1000 \times 10^{-3}) \times 0.68 \times 87.53}{1} \cong 59.534 (kW)$$

$$N = \frac{\gamma(kN/m^3) \times Q(m^3/s) \times H_m(m)}{75\eta} (BB)$$

$$N = \frac{\gamma(kN/m^3) \times Q(m^3/s) \times H_m(m)}{75\eta} \text{ (BB)}$$

$$N = \frac{1000 \times 0.68 \times 87.53}{75} = 793.8 \text{ BB}$$

Problem 7.9: The fluid flows from reservoirs (1) and (2) to valve at M. The Manning roughness coefficient for the pipes are n=0.014.

a) The piezometer height at M is 112 m,

b) And when the valve is opened at M determine the discharge.

c) Draw the energy grade line for both cases.

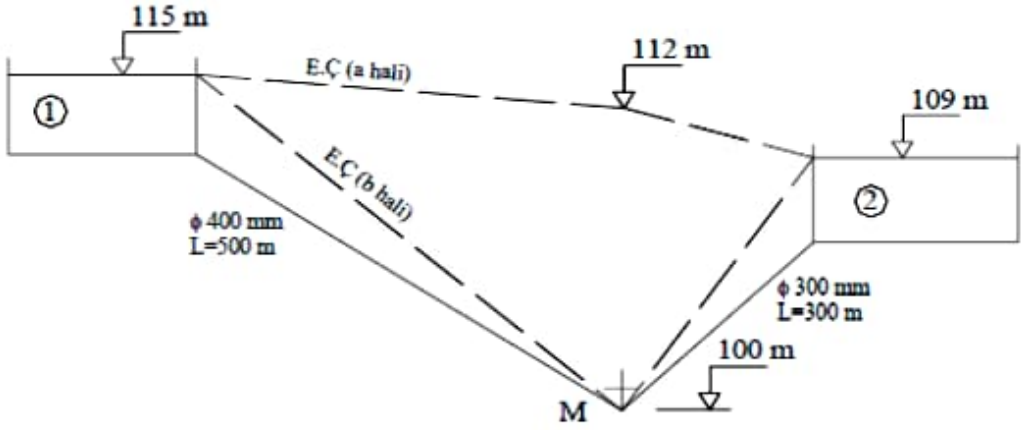


Figure 7.9

a) J_1=(115-112)/500=0.006

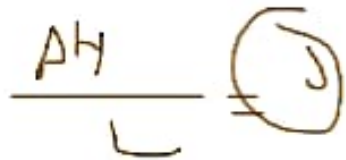
$$Q_1 = \frac{1}{n} R_1^{2/3} J_1^{1/2} A_1 = \frac{1}{0.014}\left(\frac{0.4}{4}\right)^{2/3} (0.006)^{1/2}\left(\frac{\pi \times 0.4^2}{4}\right) = 0.15 \text{ m}^3/\text{s}$$

J_2=(112-109)/300=0.01

$$Q_1 = \frac{1}{0.014}\left(\frac{0.3}{4}\right)^{2/3} (0.01)^{1/2}\left(\frac{\pi \times 0.3^2}{4}\right) = 0.09 \text{ m}^3/\text{s}$$

$Q_m = Q_1 - Q_2$

Q_m – 0.15-0.09 – 0.06 m^3/s



b) J_2=(115-100)/500=0.03

When the valve is opened the pressure head will be zero and hence the discharges will be maximum at pipes (1) and (2).

$$Q_{1max} = \frac{1}{0.014}\left(\frac{0.4}{4}\right)^{2/3}(0.03)^{1/2}\frac{\pi \times 0.4^2}{4} = 0.335\ m^3/s$$

J_2=(109-100)/300=0.03

$$Q_{2max} = \frac{1}{0.014}\left(\frac{0.3}{4}\right)^{2/3}(0.03)^{1/2}\frac{\pi \times 0.3^2}{4} = 0.156\ m^3/s$$

$Q_{Mmax}= Q_{1max}+ Q_{2max}$=0.335+0.156=0.491 m^3/s

Problem 7.10: Determine the discharge at each pipe when the friction head loss in the pipe AD is 14 m in Fig 7.10 and find the elevation of the reservoir E, draw the energy grade line of the system.

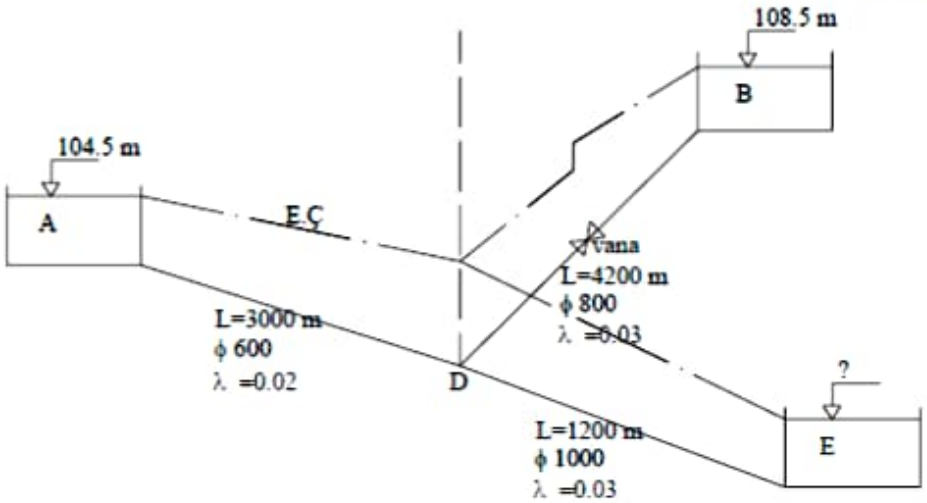


Figure 7.10

Solution:

$$\Delta h_{L_{AD}} = \lambda_{AD} \frac{L_{AD}}{D_{AD}} \frac{V_{AD}^2}{2g}$$

$$14 = 0.02 \times \frac{3000}{0.6} \times \frac{V_{AD}^2}{19.62} \qquad V_{AD} = 1.66 \text{m/s}$$

$$Q_{AD} = V_{AD}A_{AD} = 1.66 \times \frac{\pi \times 0.6^2}{4} = 0.469\ \mathrm{m^3/s}$$

$$\left(\frac{P}{\gamma} - z\right)_D = 104.5 - 14 = 90.5\ \mathrm{m}$$

so the fluid flows from reservoirs A and B to E.

$$\Delta h_{L_{BD}} = 108.5 - 905 = 18\ m$$

$$18 = 0.03 \times \frac{4200}{0.8} \times \frac{V_{BD}^2}{19.62} \qquad V_{BD} = 1.5\ m/s$$

$$Q_{BD} = V_{BD} A_{BD} = 1.5 \times \frac{\pi \times 0.8^2}{4} = 0.754\ m^3/s$$

$$Q_{DE} = Q_{AD} + Q_{BD} = 0.469 + 0.754 = 1.223\ m^3/s$$

$$1.223 = V_{DE} \frac{\pi \times 1^2}{4} \qquad V_{DE} = 1.557 m/s$$

$$\Delta h_{L_{DE}} = 0.03 \times \frac{1200}{1} \times \frac{1.557^2}{19.62} = 4.45\ m$$

$$z_E = \left(\frac{P}{\gamma} + z\right)_D - \Delta h_{L_{DE}} = 90.50 - 4.45 = 86.05\ m$$

Problem 7.11: Reservoir A and B are feeding reservoir C by a pump located at point D. The discharges from Reservoir A and B to Reservoir C have same discharges as 100 lt/sec. If the efficiency of the pump is $\eta = 0.70$, what will be required power of this pump. Determine the elevation of Reservoir B. Draw the energy grade line of the system. (λ=0.02, η=0.70, L_1=500m, ϕ_1=400mm, L_2=1000m, ϕ_2=300mm, L_3=1500m, ϕ_3=500mm.)

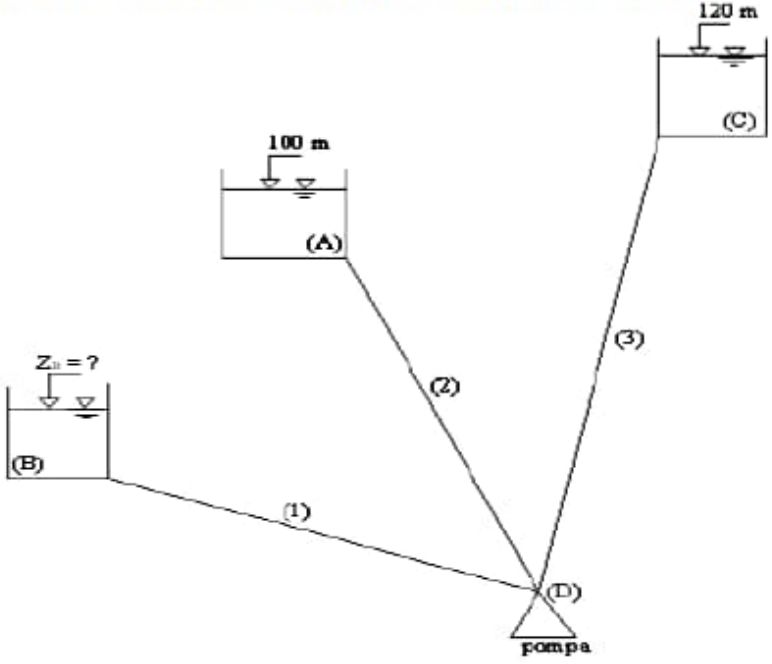


Figure 7.11

$$Q = VA$$

$$0.1 = V_2 \pi \frac{0.3^2}{4} \rightarrow V_2 = 1.41 m/s$$

$$0.1 = V_1 \pi \frac{0.4^2}{4} \rightarrow V_1 = 0.8 m/s$$

$$0.2 = V_3 \pi \frac{0.5^2}{4} \rightarrow V_3 = 1.02 m/s$$

$$\frac{P_A}{\gamma} + \frac{V_A^2}{2g} + z_A + H_m = \frac{P_C}{\gamma} + \frac{V_C^2}{2g} + z_C + \lambda \frac{L_2}{D_2} \frac{V_2^2}{2g} + \lambda \frac{L_3}{D_3} \frac{V_3^2}{2g}$$

$$100 + H_m = 120 + \lambda \frac{1000}{0.3} \frac{V_2^2}{2g} + \lambda \frac{1500}{0.5} \frac{V_3^2}{2g}$$

$$H_m = 20 + 0.02 \frac{1000}{0.3} \frac{1.41^2}{2g} + 0.02 \frac{1500}{0.5} \frac{1.02^2}{2g}$$

$$H_m = 29.94 m$$

$$\frac{P_B}{\gamma}+\frac{V_B^2}{2g}+z_B+H_m=\frac{P_C}{\gamma}+\frac{V_C^2}{2g}+z_C+\Delta H_L$$

$$29.94+z_B=120+0.02\frac{500}{0.4}\frac{0.8^2}{2g}+0.02\frac{1500}{0.5}\frac{1.02^2}{2g}$$

$$z_B=94.06m$$

$$N=\frac{\gamma Q H_m}{\eta}=\frac{10*0.2*29.94}{0.7}=85.54kw$$

Problem 7.12: Draw the energy grade lines of the systems below since the flow directions are known.

Solution: The energy grade lines have shown on the systems with dotted lines.

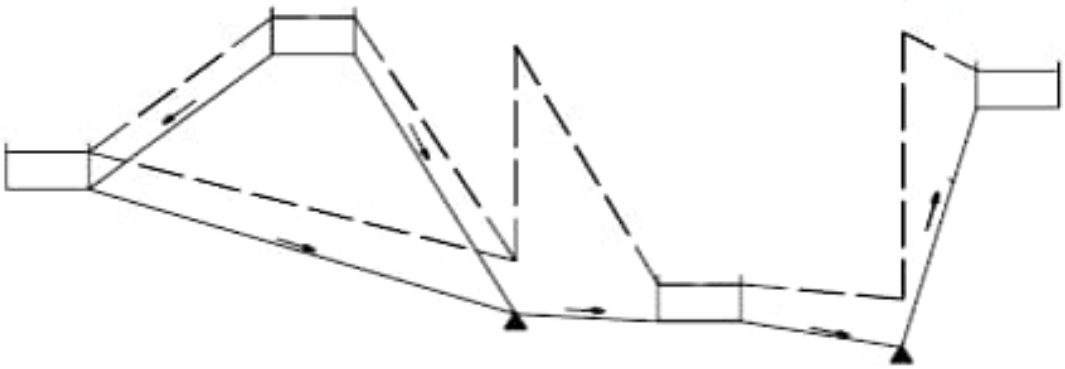


Figure 7.12a

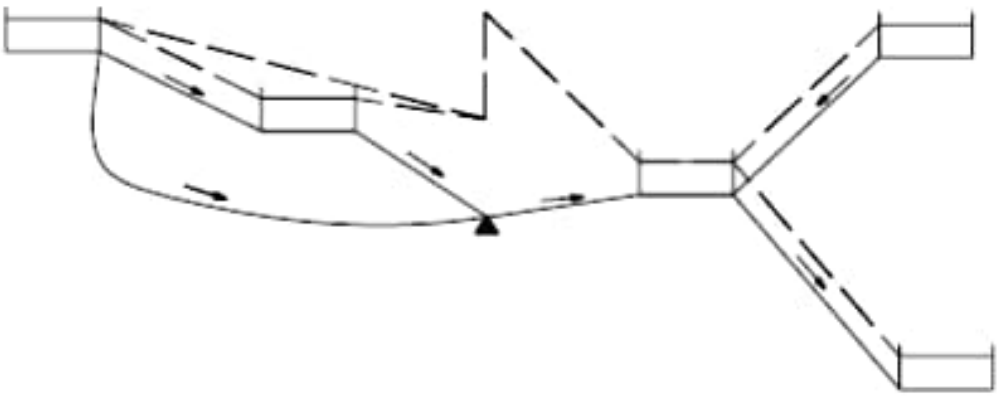


Figure 7.12b

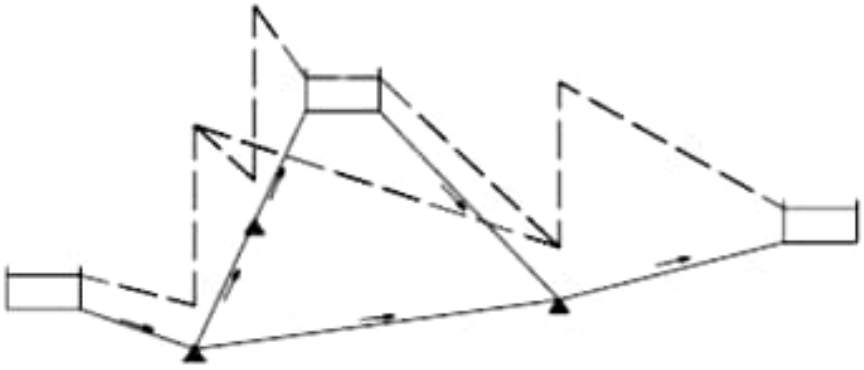


Figure 7.12c

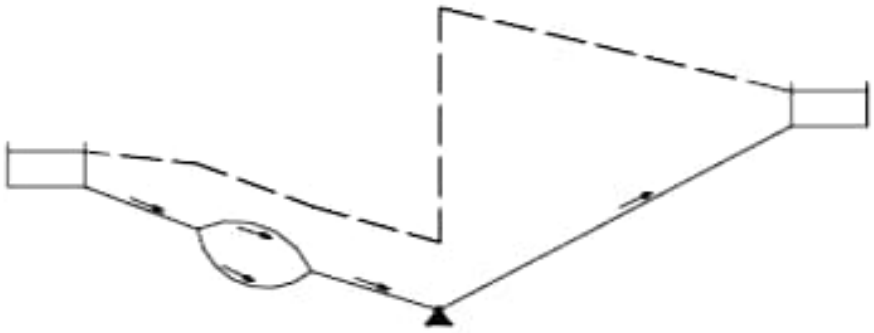


Figure 7.12d

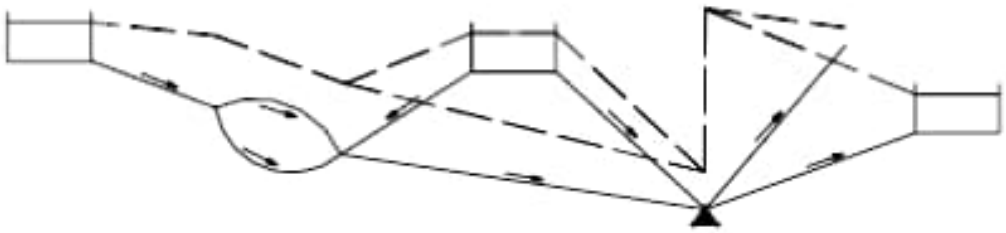


Figure 7.12e

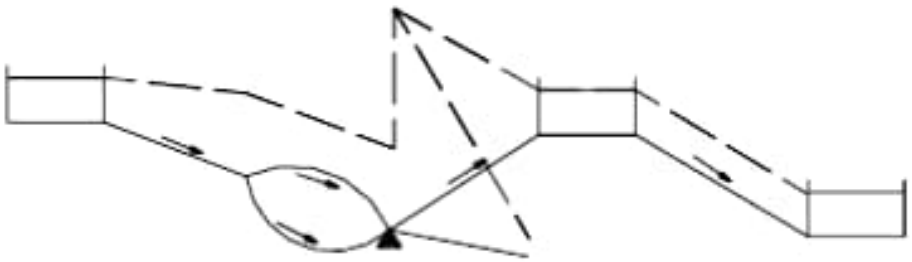


Figure 7.12f

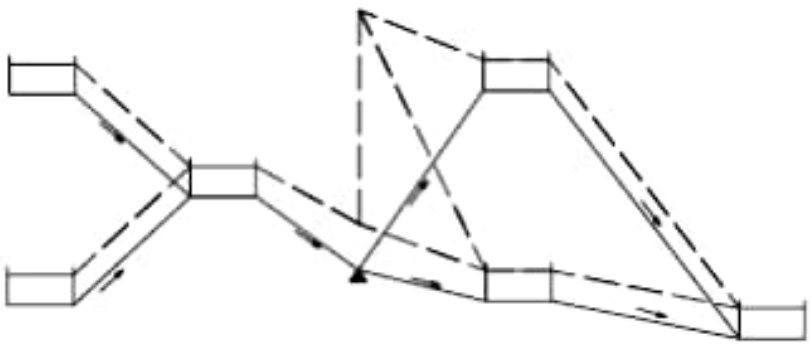


Figure 7.12g

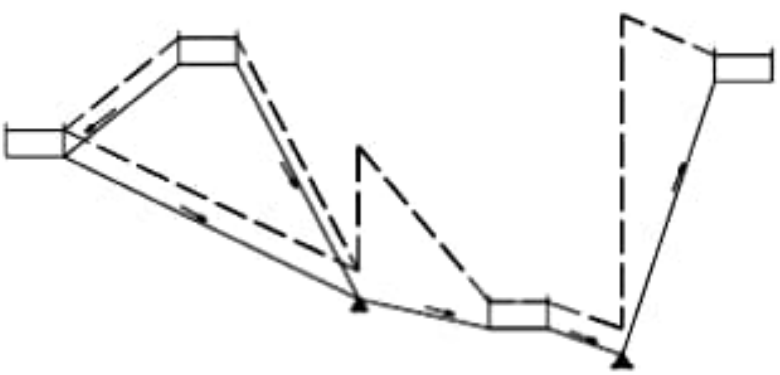


Figure 7.12h

Problem 7.13: Water flows through the pipe system as seen in Figure below. If the manometric height of the turbine is 116 m, determine the discharge in the pipe system. (Neglect head loss at the exit of the tank). ($\nu=10^{-6}$ m^2/s)

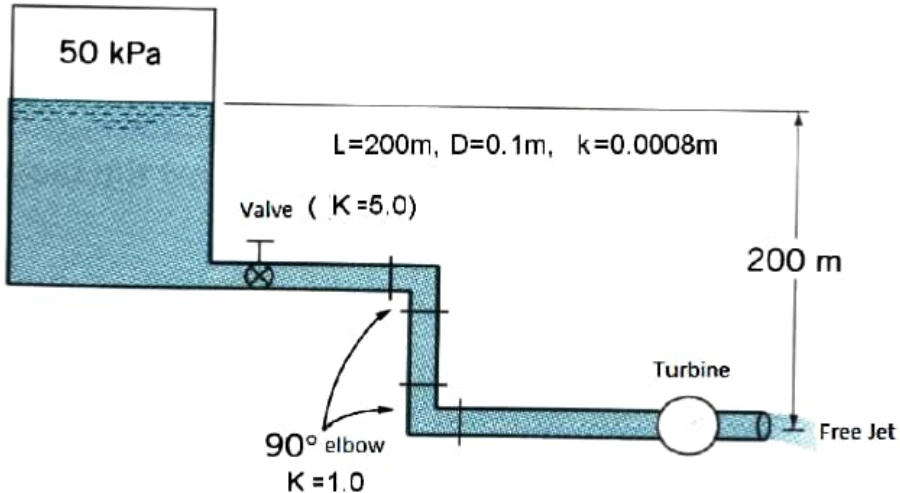


Figure 7.13

$$\frac{P_1}{\gamma}+\frac{V_1^2}{2g}+z_1-H_m=\frac{P_2}{\gamma}+\frac{V_2^2}{2g}+z_2+\sum\Delta h$$

$$\frac{50}{100}+200-116=\sum\Delta h+\frac{V^2}{2g}$$

$$89=\frac{V_2^2}{2g}+5\frac{V_1^2}{2g}+2*1*\frac{V_1^2}{2g}+\lambda\frac{200}{0.1}\frac{V_1^2}{2g}$$

$$V_2=V_1$$

$$89=(8+0.0358\frac{200}{0.1})\frac{V_1^2}{2g}$$

$$89 = 79.6\frac{V_1^2}{2g}$$

$$V_1 = 4.68 \quad m/s$$

$$Q = 4.68\frac{\pi 0.1^2}{4} = 0.0367 \quad m^3/s$$

KONTROL :

$$\mathrm{Re}_1 = \frac{4.68*0.1}{10^{-6}} = 468\,000 \cong 0.5*10^6$$

$$\frac{k}{D} = \frac{0.0008}{0.1} = 0.008$$

Hidrolik pürüzlü bölge $\lambda = 0.0358$